Elrace Mobile App — Notifications Catalogue and Trigger Points

Module-wise inventory, backend triggers, app display and tap behaviour, and issues

15 October 2026

Scope. Every notification the Elrace mobile app can show, from three sources:

- **Odoo backend** (`odoo14-clients`, branch `elrace-addons`): FCM pushes and the notification inbox.
- **Firebase Cloud Functions** (`functions/`, `functions-assignment/` in the mobile repo): chat, notes, signed documents, tasks and tickets, and announcement fan-out.
- **The app itself** (`Elrace-mobileapp-operations`, Flutter): locally scheduled notifications (prayer, tasks, check-in reminders).

Line numbers refer to the code as of 15 October 2026.

1. Summary

Item	Count / finding
Live push triggers in Odoo	About 35, across 11 business areas
Cloud Function push types	6 (chat, document signed, note shared, task, ticket, announcement/circular)
Local notifications scheduled by the app	7 active types (prayer x2, task x5); check-in/out reminders are disabled
Push types that open a screen when tapped	**3 only**: chat, task/ticket, circular/announcement
Push types that do nothing when tapped	All Odoo approval, HR request, project, attendance, document expiry, share, widget, admin and note pushes
Notification Center row tap	**Does nothing**
Confirmed duplicate	Approving an LPO, invoice or petty cash from the app sends two pushes to the requester and two to the next reviewer
Confirmed tap bug	After logging in from the sign-in screen, all notification taps are ignored until the app is restarted

The four highest-impact problems are in section 9.1.

2. How a notification reaches the phone

2.1 Odoo push path (most notifications)

1. A business event runs in Odoo (for example, a leave request is submitted, or an LPO tier is approved).

2. The code calls `elrace.notification._send_fcm_push(token, title, body, data)` (`elrace_dashboard/models/elrace_notification.py:2603`).
3. That method:
   - finds the user for the token;
   - checks the user's mute settings (`mobile.notification.preference`); if the category is muted, **nothing is sent and nothing is stored**;
   - stores an inbox row (`expo.notifications` + `expo.notification.users`) unless the record is one of the four "push-only" approval models (`employee.requests`, `purchase.order`, `account.move`, `hr.expense.sheet`);
   - sends one FCM HTTP v1 message to the user's `res.users.expo_token`.
4. FCM credentials are read from `operating.unit` record **id 6** (hard-coded).

The data sent to the phone is always only these keys: `record_id`, `model`, `status`, `category`, `click_action` (`OPEN_RECORD`), and `notification_id` / `notification_user_id` when an inbox row was stored. Any other key (for example `type`, `request_id`, `vendor_name`) is dropped from the push. A missing key arrives as the literal text `"None"`.

2.2 Pending-reviewer path (approvals)

When someone must approve something, `_notify_pending_reviewer` (`elrace_notification.py:2290`) sends three things:

- the FCM push above;
- an inbox row, **always** (it ignores the push-only rule and the mute setting);
- a push to the Elrace Hub web app (`POST {hub}/api/hub/notifications/push`).

2.3 Cloud Function path

Function	File	Fires on
`onNewChatMessage`	`functions/index.js:58`	New Firestore chat message
Note shared	`functions/notes_share_notify.js:62`	A note is shared with a user
Document signed	`functions/sign_document_sync.js:136`	A document sent for signing is signed
`onAssignedTodoCreated`, `onTodoCompleted`, `onAssignmentPushRequest`	`functions-assignment/index.js:130, 169, 224`	Task/ticket created, completed, or queued by the app
`onOdooAnnouncementPushJob`	`functions/odoo_announcement_push.js:101`	Odoo writes a job to Firestore `odoo_push_jobs` when a circular/announcement is published

These use Firestore tokens (`users/{uid}/fcm_tokens`), not the Odoo token. The announcement function fetches tokens from Odoo (`POST /api/internal/staff_fcm_tokens`).

2.4 What the app does with a push

App state	Handler	Behaviour
Open (foreground)	`FirebaseService` `onMessage` (`lib/firebase_service.dart:163`)	Works out the category (`category`, then `type`, then `model`). Checks mute. Chat is not bannered here. Everything else is shown as a local notification on channel `high_importance_channel`. Saved to the local inbox. Badge refreshed.
Background / killed	`_firebaseMessagingBackgroundHandler` (`lib/main.dart:100`)	The phone's system tray shows the push. The handler saves it to the local inbox.
Tapped	`_handleNotificationTap` (`lib/firebase_service.dart:778`)	Routes on `category` or `type`: prayer, chat, task/ticket, circular/announcement. **Anything else is "view only" and nothing opens.**

3. Notification catalogue by module

Column guide:

- **Trigger** — the exact event in code.
- **To** — who receives it.
- **Inbox** — whether it appears in the Notification Center list (server inbox).
- **Tap** — what opens when the user taps it.

3.1 HR — Employee requests (leave, sick, short leave, job mission, effective date, temporary permission)

#	Notification (title / body)	Trigger	To	Inbox	Tap
HR-1	"Short Leave Request", "Sick Leave Request", "Job Mission Request", "Effective Date Update Request", "Temporary Permission", "Leave Request" — "{employee} has submitted …". **Annual leave falls back to** "New Request Submitted" / "{employee} submitted a new request."	Request submitted from the app: `/api/submit_request` and `/api/hr/submit_request` → `send_employee_request_notification` (`elrace_notification.py:1695`; text at 1873-1906)	Next pending approver(s)	Yes	Nothing

#	Notification (title / body)	Trigger	To	Inbox	Tap
HR-2	“Request Approved” / “{approver} has approved your request.” — or “Request Rejected” / “… has rejected your request.”	Approver acts in the app: `/api/approve_reject_hr_request` → `send_employee_request_status_update` (`elrace_notification.py:2933`)	Requester	No (push only)	Nothing
HR-3	“New Request to Approve” / “A new request from {employee} needs your approval.”	Same call as HR-2, next approval line (`elrace_notification.py:2979`). **Also sent after a rejection** (likely bug).	Next approver	Yes	Nothing
HR-4	“New Employee Request Approval” / “{request} requires your approval.” — or, if no next approver, “Employee Request Processed” / “{request} has been fully processed.” (**also sent on rejection**)	Refusal in the Odoo web wizard (`request.refuse.reason`, `elrace_notification.py:1304-1440`)	Next approver, or creator	No	Nothing
HR-5	“New Employee Request” / “{request} requires your approval.”	`employee.requests.create` (`elrace_notification.py:1270`). **Probably never fires**: approvers are added after create.	First approver	—	—
HR-6	“New Employee Registered: {name}” / “Employee {name} has been confirmed and added to the system”	Draft employee confirmed (`draft.hr.employee.action_confirm`, `elrace_notification.py:17`)	Direct manager	Yes	Nothing

3.2 Approvals — LPO (`purchase.order`), Invoice (`account.move`), Petty cash (`hr.expense.sheet`)

#	Notification	Trigger	To	Inbox	Tap
AP-1	“New LPO Approval Request” / “New Invoice Approval Request” / “New Petty Cash Approval Request” — “{name} requires your approval.”	Record sent for approval: `request_validation()` (PO `elrace_notification.py:996`, invoice 546, petty cash 849)	First reviewer	Yes	Nothing
AP-2	“LPO Approved” / “INVOICE Approved” / “Petty Cash Approved” — “{approver} approved {name}”	Any tier approved, web or app: `validate_tier()` (PO 1042, invoice 568, petty cash 748)	Creator	No	Nothing

#	Notification	Trigger	To	Inbox	Tap
AP-3	“New LPO Approval Required” / “New INVOICE Approval Required” / “New Petty Cash Approval Required” — “Please review {name}”	Same, when more tiers remain	Next reviewer	Yes	Nothing
AP-4	PO: “LPI has been issued” / “{vendor}” (the app rewrites it to “LPO has been issued”). Invoice / petty cash: “… Fully Approved” / “{name} has been fully approved.”	Same, final tier	All reviewers	No	Nothing
AP-5	“{Purchase Order / Journal Entry / Expense Report} Approved” or “… Rejected” — “{approver} approved/rejected {name}.” Then “New Approval Required” to the next reviewer, or “… Fully Approved” to the requester.	Approve/reject **from the app**: `/api/record/tier_review` → `send_tier_review_update` (`elrace_notification.py:2781`)	Requester; next reviewer	Next reviewer only	Nothing
AP-6	“Petty Cash Request Submitted” / “{employee} submitted a petty cash request of {amount} AED.”	Petty cash submitted from the app: `/api/submit_expense` (`petty_cash_controller.py:494` → `elrace_notification.py:2890`)	Current reviewer	Yes	Nothing
AP-7	“Petty Cash Paid” / “{sheet} has been paid.”	Sheet paid (`approve_expense_sheets`, `elrace_notification.py:871`). With several sheets, every push names the **last** sheet (bug).	Reviewers	No	Nothing

Duplicate (confirmed): `ApprovalService.tier_review` (`elrace_backend_apis/services/approval_service.py:472–483`) calls `validate_tier()`, which already sends AP-2/AP-3/AP-4, then calls `send_tier_review_update`, which sends AP-5. The requester receives “LPO Approved” **and** “Purchase Order Approved”. The next reviewer receives “New LPO Approval Required” **and** “New Approval Required”.

Possible duplicate (verify): if submitting petty cash from the app also runs `request_validation()`, the reviewer gets both AP-6 and AP-1.

3.3 Projects

#	Notification	Trigger	To	Inbox	Tap
PR-1	“New Project Assigned” / “You have been added to {project}”	Project created (`project.project.create`, `elrace_notification.py:66`)	Everyone on the staff list, plus **hard-coded user id 874**	Yes	Nothing
PR-2	“Project Completed” / “Project {name} is now completed”	Project status changes to Completed (`elrace_notification.py:209`)	Same as PR-1	Yes	Nothing
PR-3	“New Project Assigned” / “You have been added to {project}”	Employee added to a project staff list (`elrace_project_new_theme/models/staff_list.py:137, 163, 200`)	That employee	Yes	Nothing

These three send to **all** the user’s devices (Odoo token plus every active `mobile.device` token). Most other senders reach only the last-registered device.

3.4 Documents and shared folders

#	Notification	Trigger	To	Inbox	Tap
DOC-1	“Document expiring soon” / “{document} expires on {date}. Please renew it.” (several: “You have {n} documents expiring soon …”)	**Daily cron** “Mobile: Document Expiry Soon Notifications” (`emp_mobile_conf/models/document_expiry_notify.py:18`). Documents expiring within 30 days. **Repeats every day** until expiry.	The employee	Yes	Nothing
DOC-2	“Shared Folder Update” / “New user has been added to the shared folder ‘{name}’.” — or “A user has been removed …”	`/api/cloud/folder/add_user`, `/remove_user` (`elrace_backend_apis/controllers/cloud_controller.py:308, 355`)	All folder members, including the actor	Yes	Nothing
DOC-3	“Document signed”	A document sent for signing is signed (Cloud Function `sign_document_sync.js:136`)	Sender	Yes	Opens a chat called “Document signed” (wrong). **Not shown at all while the app is open** (it carries `chat_id`, so it is treated as chat).
DOC-4	Note shared	A note is shared (`functions/notes_share_notify.js:62`)	Recipient	Yes	Nothing (no notes route)

Project document expiry (`om_project_document`) is **email only** and does not reach the app.

3.5 Announcements and circulars

#	Notification	Trigger	To	Inbox	Tap
ANN-1	“Circular published” / “Announcement published” — body is the description or text (180 characters)	Announcement made active in Odoo (`odx.announcement.write` , `elrace_notification.py:3182`) → Firestore job → Cloud Function `odoo_announcement_push.js`	All active users (general), or the selected users	Stored locally, but **hidden** from the Notification Center (still counted in the bell badge)	Opens Circulars & Announcements on the right tab and record

3.6 Attendance, face enrolment and timesheet

#	Notification	Trigger	To	Inbox	Tap
ATT-1	“Attendance Updated” / “Your attendance has been synced from BioTime. Please update the Project location.”	New attendance from a **manual** BioTime pull (`cubes_biotime/models/attendance_sync.py:1074`). The automatic Lambda crons do **not** push; the live-pull cron is inactive.	The employee	Yes	Nothing (refreshes today’s attendance status)
FACE-1	“Face enrollment verified” / “{labor} (File ID: {id}) is ready for attendance matching. Open Site Management to sync if needed.”	A labourer’s face enrolment becomes ready (`elrace_backend_apis/models/user_face_image.py:12` , processed by a 15-minute cron)	The labourer’s foreman	Yes	Nothing (push data is `"None"` for all routing keys)

There are no timesheet submit/approve pushes. Check-in/check-out reminders are local and disabled (section 4).

3.7 App configuration and admin

#	Notification	Trigger	To	Inbox	Tap
CFG-1	“Your app menu changed” / “Your dashboard widgets were updated. Please re-login to see the changes.”	An employee’s custom widget list is changed (`emp_mobile_conf/models/hr_employee.py:164` → `widget_update_notify.py:25`). **Role template changes do not push**; they force a logout instead.	The employee	Yes	Nothing (no forced re-login)

#	Notification	Trigger	To	Inbox	Tap
ADM-1	Admin-written title and message	“Send” on **Mobile Admin Notification** (`emp_mobile_conf/models/mobile_admin_notification.py:89`)	All non-labour staff, or selected employees	Yes	Nothing

3.8 Chat, tasks and tickets (Cloud Functions + app)

#	Notification	Trigger	To	Inbox	Tap
CH-1	Sender name (DM) or “{group} • {sender}” / message text, or 📷 Photo, 📎 File, 🎵 Voice message, 🎬 Video, 📝 Document for signing	New chat message. Background: Cloud Function `onNewChatMessage`. Foreground: the app’s Firestore listener (`lib/chat/services/chat_notification_service.dart:124`), which skips own messages, the open chat, muted chats, and messages older than 10 seconds.	Chat members	Yes (even while that chat is open)	Opens the chat
TK-1	New task assigned	Task created and assigned (`functions-assignment/index.js:130, 224`)	Assignee	Yes	Opens task details
TK-2	Task completed	Task completed (`functions-assignment/index.js:169`)	Creator	Yes	Opens task details
TK-3	New / completed ticket	Queued by the app (`assignment_push_service.dart`)	Assignee / creator	Yes	Opens the Tasks list (not the ticket)

Task and ticket pushes ask for Android channel `task_notifications`, which the app never creates (it creates `task_notifications_channel`), so they fall back to the default channel.

4. Notifications scheduled by the app itself (local)

#	Notification	Schedule / trigger	Channel	Cancelled when	Location
L-1	“🕌 Prayer Time” / “🔔 It’s now time for {prayer} prayer” (adhan sound)	Today’s 5 prayers, scheduled at app start and nightly (WorkManager). Egyptian method, Shafi madhab, last location or Dubai. **Inexact** on Android (exact alarm permission removed).	`prayer_adhan_channel_sound_v3`	App returns to foreground, prayer muted, rescheduled. **Not on logout.**	`lib/data/services/prayer_notification_service.dart:114`

#	Notification	Schedule / trigger	Channel	Cancelled when	Location
L-2	Same text, silent (audio played in-app)	App open and a prayer is within 5 minutes (30-second timer)	`prayer_adhan_channel_silent_v1`	On tap	`prayer_notification_service.dart:73`
L-3	"📋 New Task Assigned" / "🎫 New Ticket Assigned"	Creating a task assigned to yourself (`add_task.dart:328`); self-assigned ticket (`tasks_provider.dart:418`)	`task_notifications_channel`	Muted via task/ticket	`task_notification_service.dart:89`
L-4	"⏰ Task Deadline Tomorrow" / "{task} is due in 24 hours"	Scheduled on task create, **on the creator's phone only**	same	Task completed on this phone; tasks muted. Not on edit, delete or logout.	`task_notification_service.dart:136`
L-5	"🚨 Task Due Soon!" / "{task} is due in 1 hour"	Same	same	Same	`task_notification_service.dart:163`
L-6	"✅ Task Completed" / "{task} has been completed"	Completing a task — **notifies the person who completed it**	same	—	`task_notification_service.dart:193`
L-7	"💬 New message in {task}" / "{sender}: {text}"	Sending a task comment — **notifies the sender**	same	—	`task_notification_service.dart:229`
L-8	"⏰ Check In Reminder" / "Don't forget to Check In" (08:00–09:00 every 15 min) and "⏰ Check Out Reminder" (16:00–17:00)	**Disabled** (`scheduleCheckInReminders()` returns immediately)	—	Cancelled at start, check-in/out, logout	`checkin_reminder_notification_service.dart:171, 298`

5. Web-only notifications (never reach the app)

Area	What happens	Location
Odoo dashboard bell	`elrace.notification` records + Odoo bus (categories: employee request, LPO, employee register, project open/completed, petty cash, widget update, IT custody)	`elrace_notification.py:1657, 1835`
Live counters	Pusher channel `private-user-{uid}` (notification and waiting counters). **Pusher keys and secret hard-coded.**	`elrace_realtime/models/pusher_service.py:20`
IT custody	"IT Custody Assigned" (web bell only)	`elrace_notification.py:3024`
Helpdesk	Odoo bus + `mail.activity`	`pandora_helpdesk_customization/models/helpdesk_ticket.py:28, 87`
Delegation	`mail.activity` to the approver group; hourly cron	`delegations/models/delegation.py:101`
Elrace Hub	Pending-approval pushes to the Hub web app	`elrace_notification.py:2176`

6. Notification Center, badge and mute

6.1 Screens

Item	Behaviour	Location
Notification Center	Loads up to 50 items (server + local), **marks everything read on open**, hides circulars and announcements, no category filter. Swipe to delete; “Clear all”. **Tapping a row does nothing.**	`lib/ui/presentation/Notification/notification_screen.dart:46, 72, 144`
Bell badge	“New since the Notification Center was last opened”. Polls the server **every 5 seconds** while the home header is visible.	`home_glass_app_bar.dart:99` ; `notification_storage_service.dart:732`
App icon badge	iOS only. Android does nothing.	`ios/Runner/AppDelegate.swift:47` ; `MainActivity.kt:100`
Mute settings	Per category, from Odoo categories plus local ones (chat, prayer, task, ticket). “waiting” covers all four approval models; “alert” covers weather/safety/summer. A second mute popup in the profile card uses different rules.	`notification_mute_settings_screen.dart` ; `profile_box_with_slide_animation.dart:170`

6.2 Inbox APIs (Odoo)

Route	Purpose
`POST /api/notifications`	List (limit 50) + unread count. Model `expo.notification.users` for the logged-in user.
`POST /api/notifications/read`	Mark one read
`POST /api/notifications/delete`	Delete one
`POST /api/notifications/clear`	Delete all
`POST /api/mobile/notification/categories`	Mute categories (configured in the database, no seed data)
`POST /api/mobile/notification/preferences` , `/update`	Read / set mute
`POST /api/save_expo_token`	Register the phone’s FCM token

A daily cron keeps the last 50 inbox rows per user. The legacy module `pandora_project_timesheets_apis` registers the same routes; which one runs depends on module load order.

7. Device token lifecycle

Step	Behaviour
Registration	App calls `POST /api/save_expo_token` after login, on every resume, and on token refresh. Stored in `res.users.expo_token` (one per user). Login also stores the token per device in `mobile.device.fcm_token`.
Which device gets the push	Most senders use only `expo_token`, so **only the last device used receives the push**. Projects (PR-1 to PR-3), widget update and announcements reach all devices.
Logout	**The token is not removed.** Neither the app nor the backend clears it, so a logged-out phone keeps receiving that user's pushes. Tokens are removed only when FCM reports them invalid.
Chat tokens	Separate list in Firestore `users/{uid}/fcm_tokens`, removed on chat sign-out.

8. Android and iOS setup

Item	Detail
Android channels	`high_importance_channel` (default for FCM), `chat_messages`, `task_notifications_channel`, `prayer_adhan_channel_sound_v3`, `prayer_adhan_channel_silent_v1`. Check-in/out channels are never created. `task_notifications` (used by Cloud Functions) does not exist.
Android icon	Full-colour launcher icon used as the small icon; no FCM default icon/colour in the manifest. Usually shows as a white square.
Android exact alarms	Permissions removed, so prayer and task reminders can be late.
iOS permission	Requested natively at first launch (`AppDelegate.swift:22`), before the splash.
iOS foreground	`willPresent` forces banner + sound + badge for **every** notification, while the Dart side also shows its own banner and applies mute. Risk of double banners and muted pushes still showing. **Needs a device test.**
iOS delegate	`didReceive` and `didReceiveRemoteNotification` do not call `super`; may stop tap and background handlers. **Needs a device test.**

9. Issues found

9.1 High priority

#	Issue	Where	Effect
1	**Taps ignored after login.** `markHomeReady()` is only called from the splash screen. Logging in from the sign-in screen goes straight to `HomeScreen`.	`firebase_service.dart:58`; `sign_in_screen.dart:114`	Every chat, task and circular tap is queued and never opened until the app restarts.
2	**Most notifications do nothing on tap**, and Notification Center rows do nothing on tap. The backend sends `model` + `record_id` + `click_action: OPEN_RECORD`, but the app does not route them.	`firebase_service.dart:778–811`; `notification_screen.dart:144`	Approvers cannot open the request, LPO, invoice or petty cash from the notification.
3	**Duplicate approval pushes** when approving from the app.	`approval_service.py:472–483`	Requester and next reviewer each get two notifications.
4	**Token not cleared on logout.**	App logout helpers; backend	Another user's approvals and HR notifications can reach a shared or handed-over phone.
5	**Duplicate inbox rows / badge +2** — background handler and tap handler store the same push with different ids.	`main.dart:162`; `firebase_service.dart:700`	Inbox shows the same item twice.
6	**Legacy token endpoint is open.** `/api/update/expo/token` is `auth="public"` and accepts any `user_id`.	`pandora_tier_validation_apis/controllers/main.py:13`; `pandora_model_based_notifications/controllers/main.py:13`	Anyone can redirect a user's pushes to their own phone.

9.2 Medium priority

#	Issue	Where
7	"New Request to Approve" is also sent after an HR request is **rejected**; the web refuse wizard sends "fully processed" on rejection.	`elrace_notification.py:2979, 1418`
8	Annual leave uses the generic text "New Request Submitted".	`elrace_notification.py:1873–1906`
9	"Document signed" is invisible while the app is open and opens a fake chat when tapped.	`firebase_service.dart:190–198`
10	Document expiry reminder repeats **every day** for 30 days.	`document_expiry_notify.py:18`

#	Issue	Where
11	Petty cash “Paid” names the wrong sheet when several are paid together; petty cash final approval reads `amount_total`, which does not exist on `hr.expense.sheet` (may error — verify).	`elrace_notification.py:871, 809–837`
12	Push data contains the text `"None"` for missing keys (HR-2 `record_id`, FACE-1, CFG-1).	`elrace_notification.py:2712`
13	Task/ticket channel mismatch (`task_notifications` vs `task_notifications_channel`).	`functions-assignment/index.js:47`
14	Self-notifications: comment sender and task completer notify themselves; deadline reminders only on the creator’s phone.	`task_details.dart:531, 1122`; `add_task.dart:334`
15	Circulars/announcements raise the bell badge but are hidden from the Notification Center.	`notification_screen.dart:72`
16	Badge baseline is taken before “mark all read”, so the server-driven badge stays at 0 afterwards.	`notification_screen.dart:49–52`
17	Muted users: some senders skip the inbox row, others still write it; tasks/tickets/prayer mute is phone-only, so background pushes still show.	`elrace_notification.py:2469`; `notification_service.py:28`
18	Mark-all-read always fires two requests; badge polling every 5 seconds.	`notification_api_service.dart:505`; `home_glass_app_bar.dart:99`
19	Pushes are sent inside the business transaction; a push can go out for a change that is later rolled back.	Most senders
20	Hard-coded values: user 874 (projects), `operating.unit` 6 (FCM credentials), Pusher secrets, inbox API base URL in the app.	Various

9.3 Dead or legacy code

- **Expo senders** (tokens are now FCM, so Expo rejects them): `pandora_expo_notifications` (10-minute cron), `pandora_tier_validation_apis`, `pandora_model_based_notifications` (one HTTP call per chatter message), legacy `_create_and_send` in `my_request_apis` / `my_approvals_api`. If these modules are installed they waste a call on every event. Confirm in *Apps* and uninstall.
- **Legacy FCM server-key API** (shut down by Google in 2024): `spiffy_theme_backend/models/mail_channel.py`.
- **Never called / commented out:** `backup_send_fcm_push`, `backup_validate_tier`, `send_to_employee_and_validator`, `_send_request_creation_notifications`, `_send_review_notifications`, `_send_hr_request_notifications`, `restart_validation` push, BioTime `_send_biotime_fcm`.
- **App:** check-in/out reminders (disabled), `checkUpcomingDeadlines` (6-hourly job that only logs), FCM role-topic subscription (no sender).
- **Duplicate routes:** `pandora_project_timesheets_apis` and `elrace_backend_apis` both register `/api/submit_request`, `/api/record/tier_review`, `/api/approve_reject_hr_request`, `/api/submit_expense`, `/api/save_expo_token`, the notification routes and the folder routes.

10. Recommended next steps

1. Call `markHomeReady()` after login (sign-in and UAE PASS) and `markHomeNotReady()` on logout. **App release required.**
2. Route Odoo pushes and Notification Center rows by `model` + `record_id` to the matching detail screen (HR request, LPO, invoice, petty cash, project, documents). **App release required.**
3. Remove the second push in `ApprovalService.tier_review` (backend only).
4. Clear `expo_token` and the device token on logout (backend `linked-devices/logout-current`; app `deleteToken()`).
5. Secure or remove `/api/update/expo/token` (backend only).
6. Fix HR rejection texts, the annual-leave title, petty cash “paid” naming, and send `type` in the push data (backend only).
7. Make document expiry reminders fire at 30, 7 and 1 day(s) instead of daily (backend only).
8. Uninstall the Expo modules after confirming they are installed.
9. Test on iOS devices: double banners, mute, and tap handling.

Appendix — Main source files

Backend (`odoo14-clients`)

- `elrace_dashboard/models/elrace_notification.py` — core sender and most triggers
- `elrace_backend_apis/services/notification_service.py`, `approval_service.py`, `request_service.py`, `hr_submit_service.py`, `announcement_push_queue.py`, `face_enrollment_notification_service.py`
- `elrace_backend_apis/controllers/notification_controller.py`, `auth_controller.py`, `approval_controller.py`, `petty_cash_controller.py`, `cloud_controller.py`, `staff_fcm_tokens_controller.py`
- `emp_mobile_conf/models/document_expiry_notify.py`, `mobile_admin_notification.py`, `widget_update_notify.py`, `hr_employee.py`, `mobile_device.py`
- `elrace_project_new_theme/models/staff_list.py`, `cubes_biotime/models/attendance_sync.py`, `elrace_backend_apis/models/user_face_image.py`

App (`Elrace-mobileapp-operations`)

- `lib/firebase_service.dart`, `lib/main.dart`
- `lib/core/services/notification_storage_service.dart`, `notification_api_service.dart`
- `lib/ui/presentation/Notification/notification_screen.dart`, `notification_mute_settings_screen.dart`, `notification_category_theme.dart`
- `lib/chat/services/chat_notification_service.dart`
- `lib/data/services/task_notification_service.dart`, `prayer_notification_service.dart`, `checkin_reminder_notification_service.dart`
- `functions/index.js`, `functions/notes_share_notify.js`, `functions/sign_document_sync.js`, `functions/odoo_announcement_push.js`, `functions-assignment/index.js`
- `android/app/src/main/AndroidManifest.xml`, `ios/Runner/AppDelegate.swift`